

ITA0506-COMPUTER VISION LAB EXPERIMENT

DATE :15-07-2026

EXPNO:7

AIM: Capture video from web Camera and Display the video, in slow motion and in fast motion.

Algorithm:

- Import the OpenCV library and create a webcam capture object using cv2.VideoCapture(0).
- Check whether the webcam is opened successfully, then start reading frames one by one using cap.read().
- Display each captured frame using cv2.imshow() in a window.
- For **slow motion**, increase the delay in cv2.waitKey(); for **fast motion**, decrease the delay
- Stop the loop when the user presses a key such as q, then release the camera and destroy all OpenCV windows

Code:

```
import cv2

mode = input("Enter mode (normal/slow/fast): ").lower()

if mode == "slow":
    delay = 100
elif mode == "fast":
    delay = 5
else:
    delay = 25

cap = cv2.VideoCapture(0)

if not cap.isOpened():
    print("Error: Could not open webcam.")
    exit()
```

while True:

```
    ret, frame = cap.read()
```

```
    if not ret:
```

```
        print("Error: Could not read frame.")
```

```
        break
```

```
    cv2.imshow("Webcam Video", frame)
```

```
    if cv2.waitKey(delay) & 0xFF == ord('q'):
```

```
        break
```

```
cap.release()
```

```
cv2.destroyAllWindows()
```

OUTPUT :

